

12 TET

0	1	2	3	4	5	6	7	8	9	10	11	12
A	A#	B	C	C#	D	D#	E	F	F#	G	G#	A
	Bb			Db		Eb			Gb		Ab	

31 TET

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	0
A	A≠	A#	B♭	B <i>d</i>	B	C♭ B≠	B# C <i>d</i>	C	C≠	C#	D♭	D <i>d</i>	D	D≠	D#	E♭	Ed	E	F♭	E#	F	F≠	F#	G♭	G <i>d</i>	G	G≠	G#	A♭	A <i>d</i>	A

53 TET Holdrian comma

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	0	
A	^A	^^A	vBb	Bb	^Bb	^^Bb	vvB	vB	B	^B	^^B	vC	C	^C	^^C	vvC#	vC#	C#	^C#	^^C#	vD	D	^D	^^D	vvD#	vD#	D#	^D#	^^Eb	vvE	vE	E	^E	^^E	vF	F	^F	^^F	vvF#	vF#	F#	^F#	^^F#	vG	G	^G	^^G	vvG#	vG#	G#	^G#	^^A	vA	A
-53	-52	-51	-50	-49	-48	-47	-46	-45	-44	-43	-42	-41	-40	-39	-38	-37	-36	-35	-34	-33	-32	-31	-30	-29	-28	-27	-26	-25	-24	-23	-22	-21	-20	-19	-18	-17	-16	-15	-14	-13	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0	
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	
53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	
106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	
159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	

C	C#	D	D#	E	F	F#	G	G#	A	Bb	B
0	5	4	5	4	4	5	4	5	4	4	5

C	^C	^^C	vvC#	vC#	C#	^C#	^^C#	vD	D	^D	^^D	vvD#	vD#	D#	^^Eb	vvE	vE	E	^E	^^E	vF	F	^F	^^F	vvF#	vF#	F#	^F#	^^F#	vG	G	^G	^^G	vvG#	vG#	G#	^G#	^^A	vA	A	^A	^^A	vBb	Bb	^Bb	^^Bb	vvB	vB	B	^B	^^B	vC
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HCS Distances	1st	2nd	3rd	4th	5th	6th	7th	8ve
Major	0	9	9	4	9	9	9	4
	Cmaj7	Dm7	Em7	Fmaj7	G7	Am7	Bø7	Cmaj7
Neutral	0	8	7	8	9	8	7	7
	CNN7	vDnn7	^^EbNMN7	FvMN7	GNn7	vAm7	^^BbNN7	CNN7
SubMinor	0	9	3	10	9	3	10	9
	Csm7	Dø7	vvD#Smaj_S7	Fsm_sm7	Gm7	vvG#Smaj_SM7	Bbmaj_sm7	Csm7
H_3rd H_7th	0	9	8	5	9	8	4	10
	Cmaj_sm7	Dm7	vEø7	Fmaj7*	Gsm7	vAm7	vBbmaj7	Cmaj_sm7
SuperMajor	0	10	9	3	9	10	9	3
	CSmaj_sm7	^Dsm_sm7	^Esm_7	Fmaj_S7	Gsm7	Am7	vBbSmaj_S7	CSmaj_sm7
Major_v2	0	9	9	5	8	8	9	5
	Cmaj7*	Dm7	Em7	^FNmaj7*	G7*	vAm7	vBm7	smallFont
Neutral	0	9	6	7	9	7	7	8
	CNn7	Dø7	^^EbNMNM7	FNMN7	Gnm7	vvANN7	^^BbvMN7	CNn7

P 1	^ 1	^^ 1	v m 2	m 2	^ m 2	^^ m 2	v v M 2	v M 2	M 2	^ M 2	v v m 3	v m 3	m 3	^ m 3	^^ m 3	v v M 3	v M 3	M 3	^ M 3	^^ M 3	v 4	P 4	^ 4	^^ 4	v d 5	d 5	^ d 5	^^ d 5	v v 5	v 5	P 5	^ 5	v v m 6	v m 6	m 6	^ m 6	^^ m 6	v v M 6	v M 6	M 6	^ M 6	v v m 7	v m 7	m 7	^ m 7	^^ m 7	v v M 7	v M 7	M 7	^ M 7	^^ M 7	v 8
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52

	Interval	sub-minor	minor	neutral	major	super-major
Third		11	13	16	18	20
Seventh		42	44	46	49	51
		Diminished		Perfect		Aug
Fifth		27		31		35

How to name the chords.

Third	Fifth	Seventh	Name
supermajor	perfect	supermajor	S^S^7
supermajor	perfect	upmajor	S^^7
supermajor	perfect	major	S^M7
supermajor	perfect	downmajor	S^^7
supermajor	perfect	neutral-major	S^NM7
supermajor	perfect	neutral	S^N7
supermajor	perfect	neutral-minor	S^Nm7
supermajor	perfect	minor	S^m7
supermajor	perfect	downminor	S^vm7
supermajor	perfect	subminor	S^sm7
upmajor	perfect	supermajor	^S^7
upmajor	perfect	upmajor	^^7
upmajor	perfect	major	^M7
upmajor	perfect	downmajor	^v7
upmajor	perfect	neutral-major	^NM7
upmajor	perfect	neutral	^N7
upmajor	perfect	neutral-minor	^Nm7
upmajor	perfect	minor	^m7

upmajor	perfect	downminor	^vm7
upmajor	perfect	subminor	^sm7
major	perfect	supermajor	S^7
major	perfect	upmajor	^M7
major	perfect	major	maj7
major	perfect	downmajor	vM7
major	perfect	neutral-major	NM7
major	perfect	neutral	N7
major	perfect	neutral-minor	Nm7
major	perfect	minor	7
major	perfect	downminor	vm7
major	perfect	subminor	sm7
downmajor	perfect	supermajor	vMS^7
downmajor	perfect	upmajor	vM^M7
downmajor	perfect	major	vMM7
downmajor	perfect	downmajor	vMvM7
downmajor	perfect	neutral-major	vMNM7
downmajor	perfect	neutral	vMN7
downmajor	perfect	neutral-minor	vMNm7
downmajor	perfect	minor	vMm7
downmajor	perfect	downminor	vMvm7
downmajor	perfect	subminor	vMsm7
neutral-major	perfect	supermajor	NMS^7
neutral-major	perfect	upmajor	NM^7
neutral-major	perfect	major	NMM7
neutral-major	perfect	downmajor	NMvM7

neutral-major	perfect	neutral-major	NMNM7
neutral-major	perfect	neutral	NMN7
neutral-major	perfect	neutral-minor	NMNM7
neutral-major	perfect	minor	NMm7
neutral-major	perfect	downminor	NMvm7
neutral-major	perfect	subminor	NMsm7
neutral	perfect	supermajor	NS^7
neutral	perfect	upmajor	N^7
neutral	perfect	major	NM7
neutral	perfect	downmajor	NvM7
neutral	perfect	neutral-major	NNM7
neutral	perfect	neutral	NN7
neutral	perfect	neutral-minor	NNm7
neutral	perfect	minor	Nm7
neutral	perfect	downminor	Nvm7
neutral	perfect	subminor	Nsm7
neutral-minor	perfect	supermajor	NmS^7
neutral-minor	perfect	upmajor	Nm^7
neutral-minor	perfect	major	NmM7
neutral-minor	perfect	downmajor	NmvM7
neutral-minor	perfect	neutral-major	NmNM7
neutral-minor	perfect	neutral	NmN7
neutral-minor	perfect	neutral-minor	NmNm7
neutral-minor	perfect	minor	Nmm7
neutral-minor	perfect	downminor	Nmvm7
neutral-minor	perfect	subminor	Nmsm7

minor	perfect	supermajor	mS^7
minor	perfect	upmajor	m^7
minor	perfect	major	mM7
minor	perfect	downmajor	mvM7
minor	perfect	neutral-major	mNM7
minor	perfect	neutral	mN7
minor	perfect	neutral-minor	mNm7
minor	perfect	minor	m7
minor	perfect	downminor	mvm7
minor	perfect	subminor	msm7
downminor	perfect	supermajor	vmS^7
downminor	perfect	upmajor	vm^7
downminor	perfect	major	vmM7
downminor	perfect	downmajor	vmvM7
downminor	perfect	neutral-major	vmNM7
downminor	perfect	neutral	vmN7
downminor	perfect	neutral-minor	vmNm7
downminor	perfect	minor	vmm7
downminor	perfect	downminor	vmvm7
downminor	perfect	subminor	vmsm7
subminor	perfect	subminor	smsm7
subminor	perfect	minor	smm7
subminor	perfect	neutral	smN7
subminor	perfect	major	smM7
subminor	perfect	super-major	smS^7
minor	diminished	subminor	smø7

minor	diminished	minor	ø7
minor	diminished	neutral	øN7
minor	diminished	major	øM7
minor	diminished	super-major	øS^7
major	augmented	subminor	M+sm7
major	augmented	minor	M+m7
major	augmented	neutral	M+N7
major	augmented	major	M+7
major	augmented	super-major	M+S^7

Intervals:

Mayor	9	9	4	9	9	9	4	53
Semaja	8	8	7	8	8	7	7	53
								0
C	C	G	Bb-2	C	Eb-1			0
	9	4	9	9	4	9		44

#	Cents	Approximate Ratios*	<u>Ups and Downs Notation</u>			<u>Solfeges</u>	
0	0.00	<u>1/1</u>	P1	unison	D	Da	Do
1	22.64	<u>81/80</u> , <u>64/63</u> , <u>50/49</u>	^1	up unison	^D	Du	Di
2	45.28	<u>49/48</u> , <u>36/35</u> , <u>33/32</u> , <u>128/125</u>	^^1, vwm2	dup unison, dudminor 2nd	^^D, vEb	Di / Fre	Daw
3	67.92	<u>25/24</u> , <u>28/27</u> , <u>22/21</u> , <u>27/26</u> , <u>26/25</u>	vwA1, vm2	dudaug 1sn, downminor 2nd	vwD#, vEb	Fro	Ro
4	90.57	<u>19/18</u> , <u>20/19</u> , <u>21/20</u> , <u>256/243</u>	vA1, m2	downaug 1sn, minor 2nd	vD#, Eb	Fra	Rih
5	113.21	<u>16/15</u> , <u>15/14</u>	A1, ^m2	aug 1sn, upminor 2nd	D#, ^Eb	Fru	Ra
6	135.85	<u>14/13</u> , <u>13/12</u> , <u>27/25</u>	^^m2	dupminor 2nd	^^Eb	Fri	Ru
7	158.49	<u>35/32</u> , <u>12/11</u> , <u>11/10</u> , <u>57/52</u> , <u>800/729</u>	vwM2	dudmajor 2nd	vwE	Re	Ruh
8	181.13	<u>10/9</u>	vM2	downmajor 2nd	vE	Ro	Reh

9	203.77	<u>9/8</u>	M2	major 2nd	E	Ra	Re
10	226.42	<u>8/7</u> , <u>256/225</u>	^M2	upmajor 2nd	^E	Ru	Ri
11	249.06	<u>15/13</u> , <u>144/125</u> , 125/108	^^M2, vvm3	dupmajor 2nd, dudminor 3rd	^^E, vF	Ri / Ne	Raw
12	271.70	<u>7/6</u> , <u>75/64</u>	vm3	downminor 3rd	vF	No	Ma
13	294.34	<u>13/11</u> , <u>19/16</u> , <u>32/27</u>	m3	minor 3rd	F	Na	Meh
14	316.98	<u>6/5</u>	^m3	upminor 3rd	^F	Nu	Me
15	339.62	<u>11/9</u> , <u>243/200</u>	^^m3	dupminor 3rd	^^F	Ni	Mu
16	362.26	<u>16/13</u> , <u>100/81</u>	vvM3	dudmajor 3rd	vvF#	Me	Muh
17	384.91	<u>5/4</u>	vM3	downmajor 3rd	vF#	Mo	Mi
18	407.55	<u>19/15</u> , <u>24/19</u> , <u>81/64</u>	M3	major 3rd	F#	Ma	Maa
19	430.19	<u>9/7</u> , 14/11	^M3	upmajor 3rd	^F#	Mu	Mo
20	452.83	<u>13/10</u> , <u>125/96</u> , <u>162/125</u>	^^M3, vv4	dupmajor 3rd, dud 4th	^^F#, vG	Mi / Fe	Maw
21	475.47	<u>21/16</u> , <u>25/19</u> , <u>675/512</u> , <u>320/243</u>	v4	down 4th	vG	Fo	Fe
22	498.11	<u>4/3</u>	P4	perfect 4th	G	Fa	Fa
23	520.75	<u>27/20</u>	^4	up 4th	^G	Fu	Fih
24	543.40	<u>11/8</u> , <u>15/11</u> , <u>26/19</u>	^^4	dup 4th	^^G	Fi / She	Fu
25	566.04	<u>18/13</u>	vvA4, vd5	dudaug 4th, downdim 5th	vvG#, vAb	Pe / Sho	Fuh
26	588.68	<u>7/5</u> , <u>45/32</u>	vA4, d5	downaug 4th, dim 5th	vG#, Ab	Po / Sha	Fi
27	611.32	<u>10/7</u> , <u>64/45</u>	A4, ^d5	aug 4th, updim 5th	G#, ^Ab	Pa / Shu	Se
28	633.96	<u>13/9</u>	^A4, ^^d5	upaug 4th, dupdim 5th	^G#, ^^Ab	Pu / Shi	Suh
29	656.60	<u>16/11</u> , <u>19/13</u> , <u>22/15</u>	vv5	dud 5th	vvA	Pi / Se	Su
30	679.25	<u>40/27</u>	v5	down 5th	vA	So	Sih
31	701.89	<u>3/2</u>	P5	perfect 5th	A	Sa	Sol
32	724.53	<u>32/21</u> , <u>38/25</u> , <u>243/160</u> , <u>1024/675</u>	^5	up 5th	^A	Su	Si
33	747.17	<u>20/13</u> , <u>192/125</u> , 125/81	^^5, vvm6	dup 5th, dudminor 6th	^^A, vBb	Si / Fle	Saw
34	769.81	<u>14/9</u> , <u>25/16</u> , 11/7	vm6	downminor 6th	vBb	Flo	Lo
35	792.45	<u>19/12</u> , <u>30/19</u> , <u>128/81</u>	m6	minor 6th	Bb	Fla	Leh
36	815.09	<u>8/5</u>	^m6	upminor 6th	^Bb	Flu	Le
37	837.74	<u>13/8</u> , <u>81/50</u>	^^m6	dupminor 6th	^^Bb	Fli	Lu
38	860.38	<u>18/11</u> , <u>400/243</u>	vvM6	dudmajor 6th	vvB	Le	Luh
39	883.02	<u>5/3</u>	vM6	downmajor 6th	vB	Lo	La
40	905.66	<u>22/13</u> , <u>27/16</u> , <u>32/19</u>	M6	major 6th	B	La	Laa

41	928.30	<u>12/7</u>	^M6	upmajor 6th	^B	Lu	Li
42	950.94	<u>26/15</u> , <u>125/72</u> , <u>216/125</u>	^^M6, vvm7	dupmajor 6th, dudminor 7th	^^B, wC	Li / The	Law
43	973.58	<u>7/4</u>	vm7	downminor 7th	vC	Tho	Ta
44	996.23	<u>16/9</u>	m7	minor 7th	C	Tha	Teh
45	1018.87	<u>9/5</u>	^m7	upminor 7th	^C	Thu	Te
46	1041.51	<u>64/35</u> , <u>11/6</u> , <u>20/11</u> , <u>729/400</u>	^^m7	dupminor 7th	^^C	Thi	Tu
47	1064.15	<u>13/7</u> , <u>24/13</u> , <u>50/27</u>	vwM7	dudmajor 7th	vwC#	Te	Tuh
48	1086.79	<u>15/8</u>	vM7	downmajor 7th	vC#	To	Ti
49	1109.43	<u>19/10</u> , <u>36/19</u> , <u>40/21</u> , <u>243/128</u>	M7	major 7th	C#	Ta	Tih
50	1132.08	<u>48/25</u> , <u>27/14</u> , <u>21/11</u> , <u>52/27</u> , <u>25/13</u>	^M7	upmajor 7th	^C#	Tu	To
51	1154.72	<u>96/49</u> , <u>35/18</u> , <u>64/33</u> , <u>125/64</u>	^^M7, vv8	dupmajor 7th, dud 8ve	^^C#, wD	Ti / De	Taw
52	1177.36	<u>160/81</u> , <u>63/32</u> , <u>49/25</u>	v8	down 8ve	vD	Do	Da
53	1200.00	<u>2/1</u>	P8	perfect 8ve	D	Da	Do

